

Bhavani Prasad Kola

Vijayawada, Andhra Pradesh, India

kolabhavaniprasad@gmail.com | +91 8309304558

linkedin.com/in/kola-bhavani-prasad | github.com/bhavani-prasad



PROFESSIONAL SUMMARY

Computer Science and Engineering graduate with strong expertise in the MERN stack (MongoDB, Express.js, React.js, Node.js) and a solid foundation in Data Structures & Algorithms and Java. Passionate about frontend and full-stack web development, with hands-on experience building scalable, production-ready applications. Seeking to leverage strong problem-solving skills and technical proficiency to contribute to innovative software solutions in a challenging development role.

EDUCATION

Lakireddy Bali Reddy College of Engineering

B.Tech in Computer Science and Engineering | CGPA: 8.37/10 (79.52%)

2022 – 2026

Mylavaram, AP

KBN Junior College

Intermediate (MPC) | 92%

2019 – 2021

Vijayawada, AP

TECHNICAL SKILLS

Programming Languages: Java, JavaScript.

Frontend Technologies: React.js, HTML5, CSS3.

Backend Technologies: Node.js, Express.js.

Databases: MongoDB, SQL.

Tools: Git.

AI / ML: Machine Learning.

Core Subjects: Data Structures & Algorithms, OOP.

PROJECTS

Real-Time Sign Language Recognition using Deep Learning

2025 – 2026

Tech: Python, TensorFlow, Keras, MediaPipe, MobileNetV2, CBAM, OpenCV

- Developed a real-time sign language recognition system using **OpenCV** and **MediaPipe** to capture live webcam input, detect hand landmarks, and extract gesture features for classification.
- Built and integrated a **TensorFlow/Keras** deep learning model for accurate gesture recognition, implementing preprocessing, confidence-based prediction, and low-latency real-time inference.
- Implemented gesture-to-text and optional text-to-speech conversion, enabling seamless communication between hearing-impaired users and non-signers through an end-to-end computer vision pipeline.

AI-Powered Smart Drone System for Precision Agriculture

2025

Tech: Python, Computer Vision, MobileNetV2, Embedded Systems, IoT

- Engineered an end-to-end insect-detection pipeline trained on **50,000+ insect images**
- Designed a precision-spray control system that reduced pesticide consumption by targeting only affected zones, improving resource efficiency significantly.
- Co-invented the system; **published as a registered patent** recognizing the technical novelty of the AI-driven spraying mechanism.

API Key Management System

2026

Tech: React.js, Node.js, Express.js, MongoDB, JWT

- Developed a secure, centralized dashboard enabling developers to store, organize, rotate, and retrieve API credentials across multiple services from a single interface.
- Implemented role-based access control, encrypted key storage, and audit logging, significantly improving credential security and reducing key-sprawl across projects.

English AI Voice Call Agent

2026

Tech: React.js, Node.js, Express.js, OpenAI GPT-4.1 Mini, Retell AI, REST APIs

- Developed a responsive React.js frontend for managing an AI voice agent supporting **English**, including agent configuration, call controls, real-time conversation display, and call history visualization.

- Built interactive dashboards and integrated REST APIs to manage call sessions, conversation logs, and agent settings, ensuring a seamless user experience across devices.
- Integrated **GPT-4.1 Mini** with **Retell AI** to enable low-latency, context-aware voice conversations using real-time speech recognition and text-to-speech.

INTERNSHIPS

Biztron Softech Ltd.
MERN Stack Intern

Jun 2024 – Jul 2024
Vijayawada, AP

- Engineered modular, reusable React components, reducing UI development time and improving cross-browser responsiveness.
- Integrated and optimized RESTful APIs, streamlining data flow between frontend and backend services and enhancing application load performance.

SmartBridge (APSCHE)
AI & ML Virtual Intern

Dec 2024 – May 2025
Remote

- Applied supervised and unsupervised learning algorithms including regression, classification, clustering, and dimensionality reduction on real-world datasets.
- Executed complete ML pipeline: data acquisition, preprocessing, feature engineering, model training, hyperparameter tuning, and evaluation.
- Delivered an end-to-end ML project demonstrating practical proficiency in model deployment workflows.

SmartBridge
Salesforce Developer Intern

May 2025 – Jul 2025
Remote

- Developed Lightning Web Components (LWC) and Apex classes to implement scalable, event-driven business logic on the Salesforce platform.
- Automated multi-step business workflows using Salesforce Flow, Process Builder, and custom triggers, reducing manual processing overhead.

ACHIEVEMENTS

- **1st Prize** – National Level 48-Hour AI Hackathon at VR Siddhartha College of Engineering for Insect Detection & AI-Powered Pesticide Spraying Drone System.
- **Published Patent** – Co-inventor on an AI-based precision pesticide spraying system recognized for its novel application of computer vision in precision agriculture.
- **Research Publication** – Paper accepted, presented, and included in proceedings at the ICICT International Conference; received official certification of presentation.

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** – Amazon Web Services
- **Salesforce Developer Certification** – SmartBridge
- **ServiceNow Certified System Administrator (CSA)** – ServiceNow
- **Smart Coder Certification** – Smart Interviews (DSA & Problem Solving)